

Variety $curve_q8$ over GF(8)

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The equation

The equation of the variety is :

$$X_0^4 + X_1^4 + X_2^4 + X_0^2 X_1^2 + X_0^2 X_2^2 + X_1^2 X_2^2 + X_0^2 X_1 X_2 + X_0 X_1^2 X_2 + X_0 X_1 X_2^2 = 0$$

(1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1)

$$X_0^4 + X_1^4 + X_2^4 + X_0^2 X_1^2 + X_0^2 X_2^2 + X_1^2 X_2^2 + X_0^2 X_1 X_2 + X_0 X_1^2 X_2 + X_0 X_1 X_2^2 = 0$$

Summary

Number of automorphisms	-1
Order of set stabilizer	504
Number of points	24
Number of lines	-1
Number of singular points	-1

Set stabilizer

The stabilizer of order 504 is generated by:

$$g_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_1$$

$$g_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}_0$$

$$g_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}_0$$

$$g_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}_0$$

$$\begin{aligned}
g_5 &= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
g_6 &= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \\
g_7 &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
\end{aligned}$$

$1,0,0,0,1,0,0,0,1,1$
 $1,0,0,0,1,0,1,1,1,0$
 $1,0,0,0,1,0,1,0,1,0$
 $1,0,0,0,0,1,0,1,0,0$
 $1,0,0,1,1,0,0,0,1,0$
 $1,0,0,1,1,1,0,0,1,0$
 $0,1,0,1,1,0,0,0,1,2$

The variety has 24 points. They are: (29, 30, 31, 32, 37, 38, 39, 40, 43, 44, 47, 48, 51, 52, 55, 56, 59, 60, 61, 62, 67, 68, 69, 70)

$0 : P_{29} = (4, 2, 1)$	$9 : P_{44} = (3, 4, 1)$	$18 : P_{61} = (4, 6, 1)$
$1 : P_{30} = (5, 2, 1)$	$10 : P_{47} = (6, 4, 1)$	$19 : P_{62} = (5, 6, 1)$
$2 : P_{31} = (6, 2, 1)$	$11 : P_{48} = (7, 4, 1)$	$20 : P_{67} = (2, 7, 1)$
$3 : P_{32} = (7, 2, 1)$	$12 : P_{51} = (2, 5, 1)$	$21 : P_{68} = (3, 7, 1)$
$4 : P_{37} = (4, 3, 1)$	$13 : P_{52} = (3, 5, 1)$	$22 : P_{69} = (4, 7, 1)$
$5 : P_{38} = (5, 3, 1)$	$14 : P_{55} = (6, 5, 1)$	$23 : P_{70} = (5, 7, 1)$
$6 : P_{39} = (6, 3, 1)$	$15 : P_{56} = (7, 5, 1)$	
$7 : P_{40} = (7, 3, 1)$	$16 : P_{59} = (2, 6, 1)$	
$8 : P_{43} = (2, 4, 1)$	$17 : P_{60} = (3, 6, 1)$	